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Seat swivel assembly with locking pin

Abstract

A swivel assembly is provided for moving a vehicle seat in an angular direction. The swivel assembly may include a lower plate, a rotating plate, a gear ring, a motor, and a locking pin assembly. The lower plate may be attached to a vehicle floor. The rotating plate may be attached to the vehicle seat. The gear ring may be fixedly attached to the rotating plate. The motor may be coupled to the gear ring and drive the gear ring to move in the angular direction to cause corresponding movement of the rotating plate and the vehicle seat. The locking pin assembly may include a locking pin that is movable relative to the lower plate between an unlocked position and a locked position. The locking pin may move into and out of engagement with the rotating plate to selectively restrict and allow movement of the rotating plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/404,627, filed on Sep. 8, 2022 and U.S. Provisional Application No. 63/269,084, filed on Mar. 9, 2022. The entire disclosures of each of the above applications are incorporated herein by reference.

FIELD

(1) The present disclosure relates to a vehicle seat assembly, and more particularly, to a seat swivel assembly with a locking pin.

BACKGROUND

(2) This section provides background information related to the present disclosure and is not necessarily prior art.

(3) Vehicles such as automobiles, for example, typically include at least one seat assembly that is movable in one or more directions (e.g., fore-aft, up-down, angular, etc.) relative to a floor of the vehicle. An occupant of the vehicle may adjust the seat's position to suit the occupant's preferences for comfort and/or functionality.

(4) A seat assembly may include a swivel assembly to adjust a position of the seat assembly in an angular direction. Some swivel assemblies are complex and allow unintentional movement during a vehicle impact event and/or during the normal operation of the swivel assembly (i.e., chucking). For example, such swivel assemblies may move from a locked position to an unlocked position during a vehicle impact event, which may cause the seat to freely move in the angular direction. In another example, components of the swivel assembly may move while the vehicle is in motion, which may cause a rattling noise that is bothersome to the occupants. Identifying and restricting the unintentional movement without adversely impacting the swivel seat assemblies' intended operation has been found to be a challenge. The present disclosure provides a swivel seat assembly with a locking pin assembly that is relatively simple and reduces or eliminates unintended movement of the swivel seat assembly.

SUMMARY

(5) This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

(6) The present disclosure provides a swivel assembly for moving a vehicle seat in an angular direction relative to a vehicle floor between a first angular position and a second angular position. The swivel assembly may include a lower plate, a rotating plate, a gear ring, a motor, and a locking pin assembly. The lower plate may be configured to be attached to the vehicle floor. The rotating plate may be configured to be attached to the vehicle seat. The gear ring may be fixedly attached to the rotating plate. The motor may be coupled to the gear ring. The motor may drive the gear ring to move in the angular direction to cause corresponding movement of the rotating plate and the vehicle seat. The locking pin assembly may include a locking pin movable relative to the lower plate between an unlocked position and a locked position. The locking pin moves into engagement with the gear ring to restrict movement of the rotating plate in the locked position and the locking pin moves out of engagement with the gear ring to allow movement of the rotating plate in the unlocked position.

(7) In some configurations of the swivel assembly of the above paragraph, the locking pin assembly includes an actuator that drives movement of the locking pin between the locked and unlocked

positions.

(8) In some configurations of the swivel assembly of either of the above paragraphs, the locking pin assembly includes a plunger movably received within a housing of the actuator, and a connecting link that transmits movement of the plunger to the locking pin.

(9) In some configurations of the swivel assembly of any of the above paragraphs, the plunger translates into and out of the housing of the actuator, and movement of the plunger causes pivoting movement of the connecting link such that the locking pin is moved between the locked position and unlocked position.

(10) In some configurations of the swivel assembly of any of the above paragraphs, the swivel assembly further includes a gear coupled to the motor and meshingly engaged with the gear ring.

(11) In some configurations of the swivel assembly of any of the above paragraphs, the gear includes outer teeth and the gear ring includes inner teeth such that the outer teeth and the inner teeth mesh together.

(12) In some configurations of the swivel assembly of any of the above paragraphs, the motor drives rotational movement of the gear, and movement of the gear causes corresponding and simultaneous rotation of the gear ring.

(13) In some configurations of the swivel assembly of any of the above paragraphs, the gear ring is positioned vertically above the rotating plate.

(14) In some configurations of the swivel assembly of any of the above paragraphs, the locking pin extends through the lower plate and at least partially through the rotating plate and the gear ring in the locked position.

(15) The present disclosure provides a swivel assembly for moving a vehicle seat in an angular direction relative to a vehicle floor between a first angular position and a second angular position. The swivel assembly may include a lower plate, rotating plate and a locking pin assembly. The lower plate may be configured to be attached to the vehicle floor. The rotating plate may be configured to be attached to the vehicle seat. The rotating plate may be movable in the angular direction relative to the lower plate to cause corresponding movement of the vehicle seat between the first and second angular positions. The locking pin assembly may include a locking pin movable relative to the lower plate between an unlocked position and a locked position. The locking pin may extend at least partially through the rotating plate to restrict movement of the rotating plate when the locking pin is in the locked position, and the locking pin may allow movement of the rotating plate when the locking pin is in the unlocked position.

(16) In some configurations of the swivel assembly of the above paragraph, the swivel assembly further includes a ball bearing ring positioned between the lower plate and the rotating plate.

(17) In some configurations of the swivel assembly of either of the above paragraphs, the swivel assembly further includes an upper plate positioned vertically above the rotating plate and fixedly attached to the lower plate.

(18) In some configurations of the swivel assembly of any of the above paragraphs, the locking pin assembly includes an actuator operable to drive movement of the locking pin between the unlocked position and the locked position.

(19) In some configurations of the swivel assembly of any of the above paragraphs, the locking pin assembly includes a plunger movably received within a housing of the actuator and a connecting link coupled to the plunger to transmit movement of the plunger to the connecting link, wherein the connecting link is coupled to the locking pin. The plunger translates into and out of the housing of the actuator, and movement of the rod causes pivoting movement of the connecting link which causes linear movement of the locking pin between the locked position and unlocked position.

(20) In some configurations of the swivel assembly of any of the above paragraphs, the swivel assembly further includes a gear ring fixedly attached to the rotating plate.

(21) In some configurations of the swivel assembly of any of the above paragraphs, the gear ring includes a plurality of inner teeth.

- (22) In some configurations of the swivel assembly of any of the above paragraphs, the swivel assembly further includes a gear having outer teeth and the outer teeth of the gear meshes with the inner teeth of the gear ring.
- (23) In some configurations of the swivel assembly of any of the above paragraphs, the swivel assembly further includes a motor operable to rotate the gear and movement of the gear drives movement of the gear ring in the angular direction.
- (24) In some configurations of the swivel assembly of any of the above paragraphs, the gear ring includes a plurality of slots spaced angularly apart from each other.
- (25) In some configurations of the swivel assembly of any of the above paragraphs, the locking pin moves into one of the slots of the gear ring to restrict movement of the rotating plate and the locking pin moves out of the slots to allow movement of the rotating plate.
- (26) Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.
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Description

DRAWINGS

- (1) The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations and are not intended to limit the scope of the present disclosure.
- (2) FIG. 1 is a top view of a vehicle with a seat assembly in a first angular position according to the principles of the present disclosure;
- (3) FIG. 2 is a top view of the vehicle with the seat assembly of FIG. 1 in a second angular position;
- (4) FIG. 3 is a side view of a swivel assembly of the seat assembly of FIG. 1;
- (5) FIG. 4 is a perspective view of the swivel assembly in the first angular position;
- (6) FIG. 5 is a perspective view of the swivel assembly in the second angular position;
- (7) FIG. 6 is an exploded view of the swivel assembly;
- (8) FIG. 7 is a cross-sectional view of the swivel assembly;
- (9) FIG. 8 is a partial exploded view of an upper plate, a gear ring, and a rotating plate of the swivel assembly;
- (10) FIG. 9 is a partial exploded view of the swivel assembly;
- (11) FIG. 10 is a perspective view of a locking pin assembly in a locked position;
- (12) FIG. 11 is a partial sectional view of the locking pin assembly in the locked position;
- (13) FIG. 12 is a perspective view of the locking pin assembly in an unlocked position;
- (14) FIG. 13 is a partial sectional view of the locking pin assembly in the unlocked position;
- (15) FIG. 14 is a top view of the swivel assembly in the first angular position and the locked position;
- (16) FIG. 15 is a top view of the swivel assembly in an intermediary angular position and the unlocked position;
- (17) FIG. 16 is a top view of the swivel assembly in the second angular position and the locked position;
- (18) FIG. 17 is a perspective view of the swivel assembly with an upper plate shown in phantom lines;
- (19) FIG. 18 is a sectional view of a swivel motor assembly;
- (20) FIG. 19 is an exploded view of another swivel assembly according to the principles of the present disclosure;
- (21) FIG. 20 is a partial perspective view of a locking pin assembly of the swivel assembly of FIG. 19, where the locking pin assembly is in a locked position;

- (22) FIG. 21 is a partial perspective view of the locking pin assembly in an unlocked position;
- (23) FIG. 22 is an exploded view of the locking pin assembly;
- (24) FIG. 23 is a partial cross-sectional view of the swivel assembly with the locking pin assembly in the locked position;
- (25) FIG. 24 is a partial cross-sectional view of the swivel assembly with the locking pin assembly in the unlocked position;
- (26) FIG. 25 is another partial cross-sectional view of swivel assembly; and
- (27) FIG. 26 is a partial perspective view of the swivel assembly with an upper plate and gear ring removed.
- (28) Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

- (29) Example embodiments will now be described more fully with reference to the accompanying drawings.
- (30) Example embodiments are provided so that this disclosure will be thorough and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.
- (31) The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.
- (32) When an element or layer is referred to as being “on,” “engaged to,” “connected to,” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.
- (33) Although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.
- (34) Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,”

“upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(35) With reference to FIGS. 1-18, a vehicle **50** is provided with a seat assembly **52**. The seat assembly **52** may include a seat **54** and a swivel assembly **56**. The swivel assembly **56** may move the seat **54** relative to a floor **57** of the vehicle **50**. More specifically, the swivel assembly **56** may move the seat **54** relative to the floor **57** in an angular direction (e.g., a rotary direction) between a first angular position (FIG. 1) and a second angular position (FIG. 2). For example, the seat **54** may rotate 180 degrees in a clockwise and/or counter-clockwise direction from the first angular position to the second angular position and vice versa. Furthermore, the swivel assembly **56** may be capable of rotating the seat **54** 360 degrees in the clockwise and/or counter-clockwise direction. In some configurations, the seat **54** may be movable to additional angular positions between the first and second angular positions.

(36) The seat **54** includes a seatback **58** and a seat bottom **60**. The seatback **58** may be movable relative to the seat bottom **60** (e.g., between reclined and upright positions via a recliner assembly; not shown) and the seat bottom **60** may be movable relative to the floor **57** of the vehicle **50** (e.g., via a fore-aft adjustment assembly; not shown).

(37) With reference to FIGS. 1-6, the swivel assembly **56** may include two or more seat plates (e.g., two or more of an upper plate **70**, rotating plate **72**, or lower plate **74**), a gear ring **62**, a swivel motor assembly **64**, a locking pin assembly **66**, and a control module **68**. The upper plate **70** is fixedly attached to the lower plate **74**, the rotating plate **72** is fixedly attached to the seat bottom **60**, and the lower plate **74** is fixedly attached to the floor **57**. Accordingly, the upper and lower plates **70**, **74** are fixed relative to the floor **57** of the vehicle **50**, and the rotating plate **72** may be movable with the seat **54** relative to the floor **57** between the first angular position (FIGS. 1 and 4) and the second angular position (FIGS. 2 and 5). More specifically, the rotating plate **72** may be attached to the seat bottom **60** via a seat pan **76**. The seat pan **76** may have a central aperture **78** and may include a first pan end **80** fixedly attached to the seat bottom **60** and a second pan end **82** fixedly attached to the rotating plate **72**. The lower plate **74** is attached to the floor **57** of the vehicle **50** using a support bracket **86**. The support bracket **86** may include a first bracket end **88** fixedly attached to the lower plate **74** and a second bracket end **90** fixedly attached to the floor **57** of the vehicle **50**.

(38) With reference to FIGS. 6-7, the upper plate **70** may include a first side **96** facing the seat bottom **60** (FIG. 5) and a second side **98** facing the rotating plate **72**. A central aperture **100** may extend through the upper plate **70**. An outer flange **102** may extend annularly about the central aperture **100**. The outer flange **102** may define a radially outer perimeter of the upper plate **70**. A distal end of the outer flange **102** may extend in a generally axial direction toward the seat bottom **60**. An inner flange **104** may extend annularly about the central aperture **100**. A distal end of the inner flange **104** may extend in a generally axial direction toward the seat bottom **60** (FIG. 5). The upper plate **70** may be formed in a variety of shapes (e.g., circular or polygonal).

(39) The first side **96** of the upper plate **70** may include an annular protrusion **106** that is disposed adjacent to (and/or defined by) an outer periphery of the inner flange **104**. The second side **98** of the upper plate **70** may include a recess **108** that extends across the annular protrusion **106** such that the annular protrusion **106** does not form a complete ring. An annular first groove **110** is formed in the second side **98** of the upper plate **70** and extends around the central aperture **100**. The first groove **110** is positioned radially between the annular protrusion **106** and the outer flange **102**.

(40) The upper plate **70** may include a plurality of apertures (e.g., a plurality of first apertures **114**, a second aperture **116**, and a third aperture **118**) extending through the upper plate **70**. Each aperture in the plurality of first apertures **114** may be positioned to extend into the annular protrusion **106** and be disposed radially outward relative to the central aperture **100**. The second aperture **116** may be positioned radially between the annular protrusion **106** and the first groove **110**. The third aperture **118** may extend into the recess **108**. The upper plate **70** may be disposed within the central aperture **78** of the seat pan **76**.

(41) With continued reference to FIGS. **6-7**, the rotating plate **72** may include a first side **120** facing the upper plate **70** and a second side **122** facing the lower plate **74**. The rotating plate **72** may further include a body **124** and one or more arms **126** extending from the body **124**. Each arm **126** may include an aperture **128** such that a mechanical fastener **130** (e.g., bolt, screw, rivets, etc.) may extend through the aperture **128**. The body **124** of the rotating plate **72** may define a recess **132** such that a sidewall **134** of the recess **132** has a shape that corresponds to the shape and contour of the outer flange **102** of the upper plate **70**. A central aperture **136** may extend through the body **124** of the rotating plate **72**.

(42) The rotating plate **72** may include a second groove **138** and a third groove **140** disposed within the recess **132**. The second groove **138** may be formed in the first side **120** of the rotating plate **72** and the third groove **140** may be formed in the second side **122** of the rotating plate **72**. The second and third grooves **138**, **140** extend annularly about the central aperture **136**. The second groove **138** is substantially aligned with the first groove **110** of the upper plate **70**. Additionally, the third groove **140** is positioned radially outward relative to the second groove **138**. In other words, the third groove **140** is disposed radially between the second groove **138** and the sidewall **134** of the rotating plate **72**. In some embodiments, the second and third grooves **138**, **140** may be radially spaced apart. In other embodiments, the second and third grooves **138**, **140** may be positioned directly adjacent to each other.

(43) The rotating plate **72** may include a plurality of apertures (e.g., a plurality of first apertures **144**, and a plurality of second apertures **146**) extending through the body **124** of the rotating plate **72**. The plurality of first apertures **144** may be positioned radially between the central aperture **136** and the second groove **138** and spaced apart annularly about the central aperture **136**. The plurality of second apertures **146** may be positioned radially between the central aperture **136** and the second groove **138** and spaced apart annularly about the central aperture **136**.

(44) As shown in FIGS. **6-8**, the gear ring **62** may include an inner diametrical surface **148** and an outer diametrical surface **150**. The inner diametrical surface **148** may include a plurality of inner teeth **152**. The outer diametrical surface **150** may include a plurality of slots **154**. The slots **154** are positioned to be aligned with the second apertures **146** of the rotating plate **72** (FIG. **8**). Additionally, a plurality of apertures **156** may extend through the gear ring **62** such that the apertures **156** are aligned with the first apertures **144** of the rotating plate **72** (FIG. **8**). The gear ring **62** may be fixedly attached to the rotating plate **72** using one or more mechanical fasteners **157** extending through the first apertures **144** of the rotating plate **72** and the apertures **156** of the gear ring **62**. As illustrated in FIG. **7**, the outer diametrical surface **150** of the gear ring **62** may be positioned adjacent to the second groove **138** of the rotating plate **72** and the inner diametrical surface **148** of the gear ring **62** may be positioned adjacent to the annular protrusion **106** of the upper plate **70**. In other words, the inner teeth **152** are positioned radially between the rotating plate **72** and the annular protrusion **106** of the upper plate **70**.

(45) With continued reference to FIGS. **6-7**, the lower plate **74** may include a first side **158** facing the rotating plate **72** and a second side **160** facing the floor **57** of the vehicle **50**. The lower plate **74** may further include a body **162** and one or more arms **164** extending from the body **162**. Each arm **164** may include an aperture **166** through which a mechanical fastener **167** (e.g., bolt, screw, rivets, etc.) may extend. The body **162** may define a recess **168** defined by a sidewall **170** that may be formed to generally mirror the sidewall **134** of the rotating plate **72**. The body **162** may include a

second recess **172** that is aligned with and abuts the annular protrusion **106** of the upper plate **70**. A central aperture **174** may extend through the body **162** of the lower plate **74**. The lower plate **74** may include an inner flange **176** that defines the central aperture **174**. The inner flange **176** may extend in the opposite direction as the inner flange **104** of the upper plate **70**. In other words, the inner flange **176** may extend toward the floor **57** of the vehicle **50** (i.e., away from the upper plate **70**).

(46) The lower plate **74** may include a fourth groove **178** that extends annularly about the central aperture **174** and is positioned radially between the second recess **172** and the sidewall **170**. The fourth groove **178** is aligned with the positioning of the third groove **140** of the rotating plate **72**.

(47) The lower plate **74** may include a plurality of apertures (e.g., a plurality of first apertures **182**, a second aperture **184**, and a third aperture **186**) extending through the body **162** of the lower plate **74**. As shown in FIG. **9**, the first apertures **182** are aligned with the first apertures **114** of the upper plate **70**. Mechanical fasteners **183** may extend through the first apertures **114** of the upper plate **70** and the first apertures **182** of the lower plate **74** to fixedly secure the upper plate **70** and lower plate **74** such that the annular protrusion **106** of the upper plate **70** abuts the second recess **172** of the lower plate **74** and the central aperture **100** of the upper plate **70** is aligned with the central aperture **174** of the lower plate **74**. The second aperture **184** may be positioned radially between the sidewall **170** and the second recess **172** and aligned with the second aperture **116** of the upper plate **70**. The third aperture **186** may be positioned adjacent to the central aperture **174** and aligned with the third aperture **118** of the upper plate **70**.

(48) With continued reference to FIGS. **6-7**, a first ball bearing ring **188** is disposed between the upper plate **70** and the rotating plate **72** at the first and second grooves **110**, **138**. A second ball bearing ring **190** is disposed between the rotating plate **72** and the lower plate **74** at the third and fourth grooves **140**, **178**. The first ball bearing ring **188** has an inner diameter **D1** and the second ball bearing ring **190** has an inner diameter **D2**. The inner diameter **D2** is greater than the inner diameter **D1**. Each of the first and second ball bearing rings **188**, **190** includes a plurality of apertures **192** that are angularly spaced apart from each other. A ball bearing **194** is partially disposed within each of the apertures **192** and rotatable within the apertures **192**. Each of the grooves **110**, **138**, **140**, **178** is shaped and contoured to the shape and contour of the ball bearings **194**. More specifically, the ball bearings **194** of the first ball bearing ring **188** are partially received in the first groove **110** and the second groove **138**. The ball bearings **194** of the second ball bearing ring **190** are partially received in the third groove **140** and the fourth groove **178** (FIG. **10**).

(49) With continued reference to FIGS. **6-7** and additional reference to FIGS. **10-13**, the locking pin assembly **66** is movable between a locked position (FIGS. **10-11**) and an unlocked position (FIGS. **12-13**). The locking pin assembly **66** may include an actuator **196**, a plunger **198**, a rod **200**, a connecting link **202**, a connecting bracket **204**, a locking pin **206**, and a locking pin housing **208**. The actuator **196** may be fixedly attached to the lower plate **74** using one or more mechanical fasteners **209**, or any other suitable fastening technique. The plunger **198** may be reciprocally received in a housing **197** of the actuator **196**. An end **212** of the plunger **198** may be attached to the rod **200**. Actuation of the actuator **196** may cause the plunger **198** to move linearly relative to the actuator housing **197** (i.e., such that the end **212** of the plunger **198** moves toward the actuator housing **197**) from the position shown in FIG. **10** (corresponding to the locked position) to the position shown in FIG. **12** (corresponding to the unlocked position).

(50) The rod **200** may have a first rod end **214** and a second rod end **216**. The first rod end **214** may be received within (or otherwise attached to) the plunger **198**. The second rod end **216** may be connected to the connecting link **202**. For example, the second rod end **216** may include a curvature (or hook portion) **218** received in an aperture in the connecting link **202** to movably connect the rod **200** to the connecting link **202**.

(51) The connecting link **202** may include a first link end **220** and a second link end **222**. The connecting bracket **204** may include a first bracket end **223** and a second bracket end **224**. The

connecting link **202** may be formed in a substantially linear shape. The connecting bracket **204** may be formed in an “L” shape. The second link end **222** of the connecting link **202** may be attached to the second bracket end **224** of the connecting bracket **204** using a mechanical fastener **225**. Additionally, the first link end **220** of the connecting link **202** may be attached to the second rod end **216**. The first bracket end **223** of the connecting bracket **204** may be coupled to the locking pin housing **208** using the locking pin **206**.

(52) The locking pin housing **208** may have a first housing end **226** and a second housing end **228**. Additionally, the locking pin housing **208** may include a cylindrical tube **230** with a pair of arms **232** extending transversely from the cylindrical tube **230**. The first housing end **226** may be fixedly attached to the lower plate **74** using one or more mechanical fasteners **233** (FIG. **6**) extending through the pair of arms **232** (FIG. **7**), or any other suitable fastening technique. A locking pin seat **234** may be disposed within the cylindrical tube **230** between the first and second housing ends **226**, **228**.

(53) As shown in FIG. **11**, the locking pin **206** and a spring **235** may be disposed within the cylindrical tube **230** of the locking pin housing **208**. The locking pin **206** may have a first pin end **236** and a second pin end **238**. The second pin end **238** may be coupled to the first bracket end **223** of the connecting bracket **204** to secure the locking pin **206** with the connecting bracket **204**. The locking pin **206** may include a flange **242** that extends radially outward and is positioned between the first pin end **236** and the second housing end **228**. The spring **235** may extend helically about the locking pin **206** and may be positioned between the flange **242** and the second housing end **228**. The spring **235** biases the locking pin **206** toward the locked position (FIG. **11**).

(54) The locking pin **206** and the locking pin housing **208** may be positioned substantially perpendicular to the upper plate **70**, rotating plate **72**, and lower plate **74**. In the locked position (FIG. **11**), the locking pin **206** may extend through (or at least partially into) the second aperture **184** of the lower plate **74**, one of the second apertures **146** of the rotating plate **72**, and one of the slots **154** of the gear ring **62**. In some embodiments, the locking pin **206** may also extend through (or partially into) the second aperture **116** of the upper plate **70** when the locking pin **206** is in the locked position.

(55) When the locking pin **206** is positioned within one of the slots **154** of the gear ring **62** (i.e., when the locking pin **206** is in the locked position), the rotating plate **72** is restricted from rotating relative to the lower plate **74**. When the locking pin **206** is moved out of the slots **154** of the gear ring **62** and the second aperture **146** of the rotating plate **72** (i.e., when the locking pin **206** is in the unlocked position, as shown in FIG. **12**), the rotating plate **72** is rotatable relative to the lower plate **74**. Accordingly, the rotating plate **72** can be locked at a predefined number of positions that correspond to the number of slots **154** in the gear ring **62**. The first angular position (FIG. **14**) and the second angular position (FIG. **16**) previously described are examples of secured positions (i.e., positions at which the locking pin **206** can be moved into the locked position). Additionally, the rotating plate **72** is rotatable to a continuous range of unsecured positions when the locking pin **206** is in the unlocked position. For example, an intermediary position shown in FIG. **15** is one unsecured position.

(56) The actuator **196** may be a solenoid, a linear motor, or any other electromechanical actuator, for example. Actuation of the actuator **196** may cause the plunger **198** to move linearly relative to the actuator housing **197** (i.e., such that the end **212** of the plunger **198** moves toward the actuator housing **197**) from the position shown in FIG. **10** (corresponding to the locked position) to the position shown in FIG. **12** (corresponding to the unlocked position). Such movement of the plunger **198** causes corresponding movement of the rod **200**, the connecting link **202**, and the locking pin **206** from the locked position to the unlocked position, as will be described in more detail below.

(57) With reference to FIGS. **6** and **17-18**, the swivel motor assembly **64** may include a motor **244**, a gear rod **246**, and a gear **248**. The motor **244** may be fixedly attached to the lower plate **74** using one or more mechanical fasteners **249**, or any other suitable fastening technique. A first rod end **250**

of the gear rod **246** may be attached to the upper plate **70** and a second rod end **252** of the gear rod **246** may be attached to the motor **244**. The gear rod **246** may extend through the third aperture **186** of the lower plate **74**, the central aperture **136** of the rotating plate **72**, the third aperture **118** of the upper plate **70**, and the motor **244**. More specifically, retaining features **254** may be disposed at each of the first and second rod ends **250**, **252**. The retaining feature **254** at the first rod end **250** is coupled to the third aperture **118** of the upper plate **70** to secure the gear rod **246** to the upper plate **70**. The retaining feature **254** at the second rod end **252** is coupled to the motor **244** to secure the gear rod **246** to the motor **244**. Operation of the motor **244** causes rotation of the gear rod **246** relative to the plates **70**, **72**, **74**.

(58) The gear **248** may be a ring-shaped or tubular-shaped member having an outer surface **256** and an inner surface **258**. The inner surface **258** defines an aperture that receives the gear rod **246** and may include a plurality of splines **262**. The splines **262** mesh with splines **264** formed on the outer diametrical surface of the gear rod **246** to rotationally fix the gear **248** to the gear rod **246**. The outer surface **256** of the gear may include a plurality of outer teeth **260**. The outer teeth **260** of the gear **248** mesh with the inner teeth **152** of the gear ring **62** (as shown in FIG. **17**).

(59) With reference to FIGS. **1-18**, the operation of the seat assembly **52** will now be described. The user may initiate movement of the seat bottom **60** via a user interface (not shown) that may include switches, buttons, knobs, etc. to control movement of the seat **54** relative to the vehicle **50**. The user interface communicates the user's desired seat movement to the control module **68** (FIGS. **1-2**).

(60) In response to the user actuating the user interface to cause rotation of the seat **54**, the control module **68** causes the locking pin assembly **66** to move the locking pin **206** from the locked position (FIGS. **10-11**) to the unlocked position (FIGS. **12-13**). Specifically, the actuator **196** may drivingly engage the plunger **198** to move the rod **200** relative to the locking pin housing **208**. The plunger **198** may slidably translate into the actuator housing **197** in a telescoping manner, thereby moving the rod **200** in a corresponding and simultaneous direction. Because the rod **200** and the connecting link **202** are attached, movement of the rod **200** causes the connecting link **202** to pivot at the second link end **222**. The second link end **222** of the connecting link **202** correspondingly lowers the connecting bracket **204** and the locking pin **206**. More specifically, the first pin end **236** lowers from the second aperture **146** of the rotating plate **72** to the second aperture **184** of the lower plate **74** to place the locking pin **206** in the unlocked position. Once the locking pin **206** is placed in the unlocked position, the locking pin assembly **66** communicates with the control module **68**.

(61) Once the locking pin **206** is in the unlocked position, the control module **68** signals the swivel motor assembly **64** to move the rotating plate **72**. The swivel motor assembly **64** moves the rotating plate **72**, thereby moving the seat bottom **60**, to the user's requested position. More specifically, the motor **244** may engage the gear rod **246** to rotate. Because the gear rod **246** is fixed to the gear **248**, rotation of the gear rod **246** correspondingly and simultaneously rotates the gear **248**. Accordingly, the outer teeth **260** of the gear **248** rotate along the inner teeth **152** of the gear ring **62** to causes corresponding and simultaneous rotation of the rotating plate **72**. Meanwhile, the first and second ball bearing rings **188**, **190** rotate within their respective grooves **110**, **138**, **140**, **178** upon movement of the rotating plate **72**. Because the rotating plate **72** is fixed to the seat bottom **60**, movement of the rotating plate **72** causes corresponding and simultaneous movement of the seat bottom **60**. Once the seat bottom **60** is in the user's desired position, the swivel motor assembly **64** communicates with the control module **68**.

(62) Lastly, the control module **68** signals the locking pin assembly **66** to return the locking pin **206** from the unlocked position (FIGS. **12-13**) to the locked position (FIGS. **10-11**). The plunger **198** may slidably translate outward from the actuator **196** in a telescoping manner, thereby moving the rod **200** in a corresponding and simultaneous direction. Because the rod **200** and the connecting link **202** are attached, movement of the rod causes the connecting link **202** to pivot at the first link end **220**. The second link end **222** of the connecting link **202** correspondingly raises the connecting

bracket **204** and the locking pin **206**. The first pin end **236** moves into one of the slots **154** of the gear ring **62** to restrict movement of the gear ring **62** and to place the locking pin in the locked position. The spring **235** pushes the locking pin **206** toward the locked position, which causes corresponding movement of the connecting link **202**, rod **200**, and plunger **198**. Accordingly, the seat bottom **60** is moved to the user's desired seat position and is secured into position.

(63) As demonstrated, the swivel assembly **56** using the swivel motor assembly **64** and the locking pin assembly **66** is operable to minimize the unintentional movement of the seat plates, including reducing chucking and reducing the likelihood of movement of the seat bottom **60** from the secured position to the unsecured position during a vehicle impact event.

(64) With reference to FIGS. **19-26**, another swivel assembly **456** is provided that may be incorporated into the seat assembly **52** instead of the swivel assembly **56**. The swivel assembly **456** may move the seat **54** relative to a floor **57** of the vehicle **50**, e.g., in an angular direction (e.g., a rotary direction) between a first angular position (FIG. **1**) and a second angular position (FIG. **2**). Apart from different features described below and/or shown in the figures, the structure and function of the swivel assembly **456** may be similar or identical to that of the swivel assembly **56** described above. Therefore, similar features may not be described again in detail.

(65) Like the swivel assembly **56**, the swivel assembly **456** may include swivel assembly **56** may include a plurality of seat plates (e.g., an upper plate **470**, a rotating plate **472**, and a lower plate **474**), a gear ring **462**, a swivel motor assembly **464**, a locking pin assembly **466**, and a control module. The structure and function of the plates **470**, **472**, **474**, gear ring **462**, swivel motor assembly **464**, and control module may be similar or identical to that of the plates **70**, **72**, **74**, gear ring **62**, swivel motor assembly **64**, and control module **68** described above.

(66) However, the particular example of the swivel assembly **456** shown in FIGS. **19-26** does not include the ball bearing rings **188**, **190** and ball bearings **194**. Instead, the swivel assembly **456** may include one or more first bearing plates **588** and one or more second bearing plates **590** (shown in FIGS. **19** and **25**). The first bearing plates **588** may be disposed between the upper plate **470** and the rotating plate **472** to facilitate rotation of the rotating plate **472** relative to the upper plate **470**. The second bearing plates **590** may be disposed between the rotating plate **472** and the lower plate **474** to facilitate rotation of the rotating plate **472** relative to the lower plate **474**. The first and second bearing plates **588**, **590** may be formed from a material having a relatively low coefficient of friction. For example, the first and second bearing plates **588**, **590** may be formed from TEFLON. The first bearing plates **588** may have a first portion **591** and a second portion **593** that is angled relative to the first portion **591**. The first portion **591** is disposed axially between the plates **470**, **472**, and the second portion **593** is disposed axially and radially between the plates **470**, **472**. In this manner, the second portion **593** of the first bearing plates **588** reduces relative lateral movement between the plates **470**, **472**.

(67) Referring now to FIGS. **20-24**, the locking pin assembly **466** is movable between a locked position (FIGS. **20** and **23**) and an unlocked position (FIGS. **121** and **24**). When the locking pin assembly **466** is in the locked position, the locking pin assembly **466** engages the rotating plate **472** and the upper plate **470** to restrict rotation of the rotating plate **472** relative to the lower plate **474** (i.e., to prevent the seat assembly **52** from rotating relative to the vehicle floor **57**). When the locking pin assembly **466** is in the unlocked position, the locking pin assembly **466** disengages the upper plate **470** and the rotating plate **472** to allow rotation of the rotating plate **472** relative to the lower plate **474** (i.e., to allow the seat assembly **52** to rotate relative to the vehicle floor **57**).

(68) The locking pin assembly **466** may include an actuator **596** (e.g., a solenoid, a linear motor, or other electromechanical actuator), a plunger **598**, a cable or rod **600**, a connecting link **602**, a connecting bracket **604**, a locking pin **606**, and a locking pin housing **608**. The actuator **596** may be fixedly attached to the lower plate **474**. The plunger **598** may be reciprocally received in a housing **597** of the actuator **596**. An end **612** of the plunger **198** may be attached to the cable **600**. Actuation of the actuator **596** may cause the plunger **598** to move linearly relative to the actuator

housing **597** (i.e., such that the end **612** of the plunger **598** moves toward the actuator housing **597**) from the position shown in FIG. **20** (corresponding to the locked position) to the position shown in FIG. **21** (corresponding to the unlocked position).

(69) One end of the cable **600** may be attached to the end **612** of the plunger **598** and another end of the cable **600** may be pivotably attached to a first end **620** of the connecting link **602**. The connecting bracket **604** may be pivotably attached to a second end **622** of the connecting link **602**. The connecting bracket **604** may be fixedly attached to the locking pin **606** (e.g., by a clip, fastener, and/or any other suitable means). An intermediate portion **621** of the connecting link **602** may include a protrusion or pin **625**. The pin **625** may be slidably received in a curved slot **627** formed in the locking pin housing **608**. As shown in FIGS. **20** and **21**, the pin **625** slides along the curved slot **627** as the locking pin assembly **466** moves between the locked and unlocked positions.

(70) The connecting bracket **604** may be generally L-shaped. That is, the connecting bracket a first leg **629** and a second leg **631** that are angled relative to each other. The first leg **629** engages an end of the locking pin **606**. The second leg **631** slidably engages the locking pin housing **608**. That is, the second leg **631** may include one or more pegs **633** that are slidably received in one or more slots **635** in the locking pin housing **608**. The pegs **633** slide along the slots **635** as the locking pin assembly **466** moves between the locked and unlocked positions. The slidable engagement between the pin **625** and slot **627** and the slidable engagement between the pegs **633** and slots **635** increase the stability and rigidity of the locking pin assembly **466**.

(71) As shown in FIG. **22**, the locking pin housing **608** may include a cylindrical tube **630** and a pair of arms **632** extending transversely from the cylindrical tube **230**. The arms **632** may be fixedly attached to the lower plate **474** (e.g., via fasteners). A locking pin seat **634** (FIGS. **23** and **24**) may be disposed within the cylindrical tube **630** between the first and second axial ends of the tube **630**.

(72) As shown in FIGS. **23** and **24**, the locking pin **606** and a spring **639** may be disposed within the cylindrical tube **630** of the locking pin housing **608**. The locking pin **606** may include a flange **642** that extends radially outward and is positioned between first and second axial ends of the locking pin **606**. The spring **639** may extend helically about the locking pin **606** and may be positioned between the flange **642** and an axial end wall **643** of the locking pin housing **608**. The spring **639** biases the locking pin **606** toward the locked position (FIG. **23**).

(73) The locking pin **606** and the locking pin housing **608** may be positioned substantially perpendicular to the upper plate **470**, rotating plate **472**, and lower plate **474**. In the locked position (FIG. **23**), the locking pin **606** may extend through (or at least partially into) an aperture **584** of the lower plate **474**, an aperture **546** of the rotating plate **472**, and an aperture **554** of the upper plate **470**. In some embodiments, the locking pin **606** may also extend through (or partially into) an aperture **516** of a guide ring **461** when the locking pin **606** is in the locked position. The guide ring **461** may be fixed relative to the rotating plate **472**. In some rotation positions of the seat assembly **52**, the locking pin **606** may also engage an aperture or slot of the gear ring **462** when in the locked position.

(74) In the unlocked position, the locking pin **606** is retracted out of the apertures **554**, **546**, **516** to disengage the locking pin **606** from the guide ring **461**, rotating plate **472**, and upper plate **470** to allow rotation of the seat assembly **52** relative to the vehicle floor **57**.

(75) The guide ring **461** can be designed to include a desired number of apertures **516** at desired locking positions to suit a given application. For example, if the vehicle manufacturer desires the swivel assembly **456** to lock a 0-degree, 90-degree, and 180-degree positions, the guide ring **461** may include only three apertures **516** positioned at the 0-degree, 90-degree, and 180-degree positions, while the rotating plate **472** and upper plate **470** may include additional apertures **546**, **554** to suit a wider variety of applications that may call for a different number of locking positions and/or locking positions at different locations. In this manner, only the guide ring **461** needs to be made according to any specific application and the plates **472**, **470** can be generic to all

applications.

(76) The term “module” or the term “controller” may be replaced with the term “circuit.” The term “module” may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor circuit (shared, dedicated, or group) that executes code; a memory circuit (shared, dedicated, or group) that stores code executed by the processor circuit; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

(77) The module may include one or more interface circuits. In some examples, the interface circuits may include wired or wireless interfaces that are connected to a local area network (LAN), the Internet, a wide area network (WAN), a controller area network (CAN), or combinations thereof. The functionality of any given module of the present disclosure may be distributed among multiple modules that are connected via interface circuits. For example, multiple modules may allow load balancing. In a further example, a server (also known as remote, or cloud) module may accomplish some functionality on behalf of a client module.

(78) The term code, as used above, may include software, firmware, and/or microcode, and may refer to programs, routines, functions, classes, data structures, and/or objects. The term shared processor circuit encompasses a single processor circuit that executes some or all code from multiple modules. The term group processor circuit encompasses a processor circuit that, in combination with additional processor circuits, executes some or all code from one or more modules. References to multiple processor circuits encompass multiple processor circuits on discrete dies, multiple processor circuits on a single die, multiple cores of a single processor circuit, multiple threads of a single processor circuit, or a combination of the above. The term shared memory circuit encompasses a single memory circuit that stores some or all code from multiple modules. The term group memory circuit encompasses a memory circuit that, in combination with additional memories, stores some or all code from one or more modules.

(79) The term memory circuit is a subset of the term computer-readable medium. The term computer-readable medium, as used herein, does not encompass transitory electrical or electromagnetic signals propagating through a medium (such as on a carrier wave); the term computer-readable medium may therefore be considered tangible and non-transitory. Non-limiting examples of a non-transitory, tangible computer-readable medium are nonvolatile memory circuits (such as a flash memory circuit, an erasable programmable read-only memory circuit, or a mask read-only memory circuit), volatile memory circuits (such as a static random access memory circuit or a dynamic random access memory circuit), magnetic storage media (such as an analog or digital magnetic tape or a hard disk drive), and optical storage media (such as a CD, a DVD, or a Blu-ray Disc).

(80) The apparatuses and methods described in this application may be partially or fully implemented by a special purpose computer created by configuring a general purpose computer to execute one or more particular functions embodied in computer programs. The functional blocks, flowchart components, and other elements described above serve as software specifications, which can be translated into the computer programs by the routine work of a skilled technician or programmer.

(81) The computer programs include processor-executable instructions that are stored on at least one non-transitory, tangible computer-readable medium. The computer programs may also include or rely on stored data. The computer programs may encompass a basic input/output system (BIOS) that interacts with hardware of the special purpose computer, device drivers that interact with particular devices of the special purpose computer, one or more operating systems, user applications, background services, background applications, etc.

(82) The computer programs may include: (i) descriptive text to be parsed, such as HTML

(hypertext markup language), XML (extensible markup language), or JSON (JavaScript Object Notation) (ii) assembly code, (iii) object code generated from source code by a compiler, (iv) source code for execution by an interpreter, (v) source code for compilation and execution by a just-in-time compiler, etc. As examples only, source code may be written using syntax from languages including C, C++, C #, Objective C, Swift, Haskell, Go, SQL, R, Lisp, Java®, Fortran, Perl, Pascal, Curl, OCaml, Javascript®, HTML5 (Hypertext Markup Language 5th revision), Ada, ASP (Active Server Pages), PHP (PHP: Hypertext Preprocessor), Scala, Eiffel, Smalltalk, Erlang, Ruby, Flash®, Visual Basic®, Lua, MATLAB, SIMULINK, and Python®.

(83) The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A swivel assembly for moving a vehicle seat in an angular direction relative to a vehicle floor between a first angular position and a second angular position, the swivel assembly comprising: a lower plate configured to be attached to the vehicle floor; a rotating plate configured to be attached to the vehicle seat; a gear ring fixedly attached to the rotating plate; a motor coupled to the gear ring, wherein the motor drives the gear ring to move in the angular direction to cause corresponding movement of the rotating plate and the vehicle seat; and a locking pin assembly including a locking pin movable relative to the lower plate between an unlocked position and a locked position, wherein in the locked position, the locking pin engages the rotating plate to restrict rotation of the rotating plate, and wherein in the unlocked position, the locking pin disengages the rotating plate to allow rotation of the rotating plate, wherein the locking pin assembly includes an actuator that drives movement of the locking pin between the locked and unlocked positions, wherein the locking pin assembly includes: a plunger movably received within a housing of the actuator, and a connecting link that transmits movement of the plunger to the locking pin, wherein the plunger translates into and out of the housing of the actuator, and movement of the plunger causes pivoting movement of the connecting link such that the locking pin is moved between the locked position and unlocked position, wherein the locking pin assembly includes a locking pin housing that reciprocatingly receives the locking pin, wherein the locking pin housing includes a curved slot that slidably receives a protrusion extending from the connecting link, wherein the locking pin assembly includes a connecting bracket that is attached to the locking pin and slidably engages the locking pin housing, and wherein the connecting bracket is pivotably attached to the connecting link.

2. The swivel assembly of claim 1, further comprising a gear coupled to the motor and meshingly engaged with the gear ring.

3. The swivel assembly of claim 2, wherein the gear includes outer teeth, and wherein the gear ring includes inner teeth that mesh with the outer teeth.

4. The swivel assembly of claim 3, wherein the motor drives rotational movement of the gear, and movement of the gear causes corresponding and simultaneous rotation of the gear ring.

5. The swivel assembly of claim 1, wherein the gear ring is positioned vertically above the rotating plate.

6. A swivel assembly for moving a vehicle seat in an angular direction relative to a vehicle floor between a first angular position and a second angular position, the swivel assembly comprising: a lower plate configured to be attached to the vehicle floor; a rotating plate configured to be attached

to the vehicle seat; a gear ring fixedly attached to the rotating plate; a motor coupled to the gear ring, wherein the motor drives the gear ring to move in the angular direction to cause corresponding movement of the rotating plate and the vehicle seat; and a locking pin assembly including a locking pin movable relative to the lower plate between an unlocked position and a locked position, wherein in the locked position, the locking pin engages the rotating plate to restrict rotation of the rotating plate, and wherein in the unlocked position, the locking pin disengages the rotating plate to allow rotation of the rotating plate, wherein the gear ring is positioned vertically above the rotating plate, and wherein the locking pin extends through the lower plate and at least partially through the rotating plate and the gear ring in the locked position, wherein the locking pin assembly includes an actuator that drives movement of the locking pin between the locked and unlocked positions, wherein the locking pin assembly includes: a plunger movably received within a housing of the actuator, and a connecting link that transmits movement of the plunger to the locking pin, wherein the locking pin assembly includes a locking pin housing that reciprocatingly receives the locking pin, wherein the locking pin housing includes a curved slot that slidably receives a protrusion extending from the connecting link, wherein the locking pin assembly includes a connecting bracket that is attached to the locking pin and slidably engages the locking pin housing, and wherein the connecting bracket is pivotably attached to the connecting link.

7. The swivel assembly of claim 6, wherein the plunger translates into and out of the housing of the actuator, and movement of the plunger causes pivoting movement of the connecting link such that the locking pin is moved between the locked position and unlocked position.

8. The swivel assembly of claim 6, further comprising a gear coupled to the motor and meshingly engaged with the gear ring.

9. The swivel assembly of claim 8, wherein the gear includes outer teeth, and wherein the gear ring includes inner teeth that mesh with the outer teeth.

10. The swivel assembly of claim 9, wherein the motor drives rotational movement of the gear, and movement of the gear causes corresponding and simultaneous rotation of the gear ring.

11. The swivel assembly of claim 6, wherein the gear ring includes a plurality of slots spaced angularly apart from each other.

12. The swivel assembly of claim 11, wherein the locking pin moves into one of the slots of the gear ring to restrict movement of the rotating plate and the locking pin moves out of the slots to allow movement of the rotating plate.
